

FULLSTACK ACADEMY Cybersecurity Capstone

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OPERATION CATCH THE HACKER

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Full Capstone Project Summary

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Attack · Phishing · Detection · Defense — One complete cycle.

MEL

RED TEAM · ATTACK

JAN

RED TEAM · PHISHING

KENNY

BLUE TEAM · DETECT

QUEEN

BLUE TEAM · DEFENSE

Date	June - July 2026	Course	Fullstack Academy Cybersecurity Capstone
Target	Kali Linux VM IP 10.211.55.9	Attacker	Kali Linux VM IP 10.211.55.8
Tools	Nmap, Hydra, Wireshark, Snort, UFW	Network	Parallels Shared Network 10.211.55.x

MISSION OVERVIEW

Operation Catch the Hacker is the Fullstack Academy Cybersecurity Capstone project for the June-July 2026 cohort. A four-person team ran a full red versus blue security exercise against an isolated lab network, covering every phase of a real attack cycle: reconnaissance and exploitation by the Red Team, and detection, hardening, and awareness training by the Blue Team.

Everything ran inside an authorized lab using two Kali Linux virtual machines on the same Parallels shared network. No real systems, no real data, and no real people were ever targeted. The goal was to demonstrate a complete attack-detect-defend cycle and produce professional documentation of every phase.

Team Member	Role	Responsibility
Melanie Seguin (Mel)	Red Team Operator	Recon, SSH brute force, Wireshark capture
Jan Lelie	Recon & Social Engineering	OSINT recon, AI-assisted phishing samples
Kenny Panyavong	Blue Team Detection	Wireshark analysis, Snort rules, tcpdump
Queen Mugoli	Blue Team Defense	UFW firewall hardening, phishing guide

HOW THE FOUR PARTS CONNECT

Each role feeds the next. Mel ran the attack and saved a full Wireshark capture, which Kenny used to find the attack in the packets and write detection rules. Jan produced three annotated phishing samples, which Queen turned into the phishing awareness guide. Queen also used Mel's attack to test whether her firewall rules successfully blocked it. The result is one continuous chain: attack, detect, defend, and educate.



OPERATION CATCH THE HACKER — Project Summary

MEL

RED TEAM · ATTACK Melanie Seguin | Attacker IP: 10.211.55.8 | Target IP: 10.211.55.9

Mel set up two Kali Linux VMs on the same Parallels network. On the target, SSH was enabled and a deliberately weak password was set to simulate a poorly secured machine. Wireshark was started on the attacker VM before any commands were run, capturing every packet as evidence.

An Nmap service version scan identified port 22 running OpenSSH 10.0p2. Hydra then ran a brute force attack using a short wordlist of common passwords. The correct credential was found in under ten seconds. The full attack — scan to confirmed access — took under one minute.

No.	Finding	Severity
1	SSH exposed on port 22 with no access restrictions or IP filtering	High
2	Weak password recovered by Hydra in under ten seconds	Critical
3	No account lockout detected — Hydra completed all attempts without interruption	High
4	OpenSSH 10.0p2 version exposed via service banner, enabling targeted exploitation	Medium

Attack result: [22][ssh] host: 10.211.55.9 login: parallels password: password123

JAN

RED TEAM · RECON & SOCIAL ENGINEERING Jan Lelie | Tools: OSINT recon, AI assistant (Claude)

Jan documented the attacker reconnaissance phase, outlining how an attacker profiles a target company before launching a phishing campaign: employee names and job titles from LinkedIn, email format from contact pages, org chart from reporting lines, and software stack from job postings.

Using the AI assistant, Jan produced three realistic phishing samples for the team to study, each using a different attack technique. Every sample was annotated with its red flags.

Sample	Technique	Key Red Flags
S1 — IT Reset	Credential harvesting	Look-alike domain, urgent threat, fake login link, misspelling
S2 — CEO Fraud	Business email compromise	Gmail CEO sender, gift card payload, secrecy directive, misspelling
S3 — Delivery	Malware attachment	Typosquat domain, 48hr deadline, .zip shipping label, misspelling

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KENNY

BLUE TEAM · DETECTION Kenny Panyavong | Tools: Wireshark, Snort, tcpdump | Capture: 2,245 packets

Kenny received Mel's capture file via Slack and opened it in Wireshark. Applying a SYN packet filter immediately revealed 2,092 packets from the attacker hitting dozens of ports — the definitive signature of an Nmap service version scan. Filtering to `tcp.port==22` isolated 181 SSH packets, confirming the brute force phase.

Inspecting individual SSHv2 packets revealed two key Hydra fingerprints: high-numbered ephemeral source ports (a new socket per attempt) and the `SSH-2.0-libssh_0.11.2` client banner — a string no legitimate user session produces. A Snort detection rule was written, installed, and confirmed to load against the full capture. `tcpdump` was used as supporting evidence.

No.	Finding	Severity
1	2,092 SYN packets from 10.211.55.8 confirm a full Nmap service version scan	High
2	181 SSH packets with multiple ephemeral ports confirm automated Hydra brute force	Critical
3	SSH-2.0-libssh_0.11.2 banner in packet data — direct fingerprint of the Hydra tool	High
4	No account lockout observed — all attempts completed without interruption	High
5	OpenSSH 10.0p2 version exposed via service banner	Medium

Snort rule: `alert tcp any any -> any 22 (msg:"SSH login attempts, possible brute force"; sid:1000001; rev:1;)`

QUEEN

BLUE TEAM · DEFENSE & HARDENING Queen Mugoli | Tools: UFW, Nmap | Contributor: Melanie Seguin

Queen configured the UFW firewall on the Kali target VM to block Mel's attack. UFW was not pre-installed on this Kali build and was added via `apt`. Once installed, two deny rules were applied: one blocking all inbound traffic from the attacker IP 10.211.55.8, and one closing port 22/tcp from any source.

Before and after Nmap scans document the state change. Queen also produced the phishing awareness guide using Jan's three annotated samples, covering all four required red flags across credential harvesting, CEO fraud, and malware delivery lures.

No.	Finding	Severity
1	UFW firewall installed and enabled on the Kali target VM	Mitigated
2	Attacker IP 10.211.55.8 blocked via deny rule — all inbound traffic stopped	Mitigated
3	Port 22/tcp denied from any source — SSH no longer accessible from the network	Mitigated
4	Before-and-after Nmap scans confirm the hardened state	Complete
5	Phishing guide built from Jan's three samples with all red flags documented	Complete

OVERALL OUTCOME

Operation Catch the Hacker achieved its objective. The Red Team demonstrated how two common misconfigurations — an SSH service exposed to the network with no restrictions and a weak password with no lockout policy — can be exploited in under one minute using only tools built into Kali Linux.

The Blue Team demonstrated that the same attack is detectable in packet data and preventable with basic firewall rules. Kenny found the attack in the capture before any written findings were shared. Queen blocked it with three UFW commands. The phishing awareness component showed how attackers use AI and public information to build convincing lures, and how to recognize them.

Phase	Owner	Outcome	Status
Attack	Mel	SSH brute force succeeded in < 1 min	Complete
Phishing	Jan	3 annotated samples + red flag guide	Complete
Detection	Kenny	Attack found in packets, Snort rule fired	Complete
Defense	Queen	Firewall blocked attack, guide delivered	Complete

Four playbooks, one team. The Red Team made it happen. The Blue Team caught it and stopped it. Operation Catch the Hacker complete.

OPERATION CATCH THE HACKER

RED TEAM · ATTACK

RED TEAM · PHISHING

BLUE TEAM · DETECTION

BLUE TEAM · DEFENSE

What began as a classroom exercise became a demonstration of something real. A team of four cybersecurity students designed, executed, detected, and shut down a complete attack cycle against a live lab environment — using only the tools built into Kali Linux and the knowledge they built over the course of this program.

Mel mapped the target, found the open door, and walked through it in under one minute. Jan showed how attackers do their homework before they ever send a single email — and how to recognize the traps they set. Kenny found every trace of the attack hiding inside 2,245 packets and wrote the rule that would have caught it. Queen closed the door, locked it, and made sure no one on the team would fall for the next one.

No real systems. No real data. No real victims. But every technique, every tool, every finding, and every lesson is real. The misconfigurations Mel exploited exist in production environments every day. The phishing lures Jan built are indistinguishable from ones landing in inboxes right now. The Snort rule Kenny wrote would fire in a real SOC. The UFW commands Queen ran would harden a real server.

**1 min Attack
to access****2,245 Packets
analyzed****3 Phishing
samples****2 Firewall
rules****4 Red flags
per sample****34 Pages of
evidence**

This is what the end of a cybersecurity program looks like. Not a test. Not a quiz. A real exercise, documented with real evidence, completed by a real team. Operation Catch the Hacker is complete.

**MEL Melanie Seguin Red
Team Operator****JAN Jan Lelie Recon &
Social Eng.****KENNY Kenny Panyavong
Detection Analyst****QUEEN Queen Mugoli
Defense & Hardening***Fullstack Academy Cybersecurity | June - July 2026 | Operation Catch the Hacker*